

When complexity turns into local prosperity: A system dynamics approach to meeting the challenges of the rural electricity-development nexus

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ARTICLE INFO

Article history:

Received 12 June 2020

Revised 24 October 2020

Accepted 27 October 2020

Available online 19 November 2020

Keywords:

Rural electrification

Electricity-development nexus

System-dynamics

Policy testing

Energy modelling

ABSTRACT

The rural electrification-based literature reports a limited modelling knowledge concerning the long-term dynamics of the nexus between electricity access and local development in rural areas of developing countries. This can hinder the achievement of the Agenda 2030 goals related to sustainable access to electricity for all. In order to overcome this literature gap and support the implementation of sustainable long-term rural electrification programmes, this paper builds upon the research on a system dynamics model on the local dimension of the electricity-development nexus developed by the authors by testing it and deriving useful guidelines for supporting future electrification actions in sub-Saharan Africa. All the steps are based on a real case-study as reference, i.e. a hydroelectric-based electrification programme implemented in the rural community of Ikondo, Tanzania, in 2005 by the Italian NGO named CEFA Onlus. The results of the model testing lead to a novel quantitative assessment of the most relevant dynamics affecting the local electricity-development nexus, and it provides a novel discussion on model results when its inputs take on different values, until the extreme ones, and as if the model were tested for different contexts than Ikondo. Policy testing is also performed for exploring model behaviour when subjected to different policies and exogenous decision-making processes. It provides a list of complementary activities to couple with electrification programmes for enhancing their positive impact on rural communities. These results can support the definition of useful guidelines and best practices for rural electrification in sub-Saharan Africa, and they advocate an updating of the traditional monitoring and evaluation frameworks commonly used for assessing energy access projects.

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Introduction

Rural electrification is expected to largely contribute to the achievement of the electricity access goal of the Agenda 2030 of the United Nations, since people still living without electricity will live predominantly in rural areas (IEA, 2018). According to this, massive financial resources have been invested and will be invested in the next decade for increasing the electrification rate in developing countries, especially in sub-Saharan Africa. In this context, the scientific research contributed to the topic of rural electrification by involving many different disciplines, and by reporting case-studies, reviews, feasibility analyses, technological assessments, policy analyses, and conceptual studies. A survey carried out by the authors on the Scopus database indicates that 1'408 documents have been published from 1984 to 2018, which include "rural electrification" in the author- or indexed-keywords. The global and scientific commitment to meeting the challenge of rural electrification led to successful achievements in terms of increased access to

electricity worldwide – the International Energy Agency (IEA) states that in 2018 and 2019 the global population without access to electricity falls below 1 billion for the first time (IEA, 2018, 2019). At the same time, by analysing the issue of electricity access beyond a simplistic binary metrics (viz. "access/no access"), a large part of the literature also reports paradoxical impacts of electricity on the life of rural communities. For example, Matinga and Annegarn (2013) report the electrification outcomes experienced in the village of Tsilitwa in South Africa, and they conclude that electrification can have negative impacts on local people, such as rising sense of income disparity. With their comprehensive review of the scientific literature regarding the nexus between rural electrification and local development, the Author (Riva, Ahlborg, et al., 2018) draw the attention of the scientific community to the complex socio-economic dynamics that can determine the successful or the failure of rural electrification programmes. Though the use of causal diagrams, they confirm that the behaviour/outcome of the nexus cannot be intuitively understood, being a complex system (Forrester, 1971).

Causal diagrams used by the Author are similar to the causal-loop diagrams used in system dynamics (SD) modelling methods. SD is a

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systems-modelling approach first developed by Prof. J.W. Forrester, Massachusetts Institute of Technology (MIT), in 1950s and formalised in his famous book entitled *Industrial Dynamics* (Forrester, 1961). It is used for understanding, describing, tackling and modelling well-defined endogenous problems concerning existing systems that are suitable to be formulated as interaction between elements through causal relationships, such as the electricity-development nexus. In SD, causal-loop diagrams are commonly used for formulating a problem through a dynamic hypothesis, for communicating a model (Morecroft, 1982), and for making qualitative analysis of complex systems (Wolstenholme & Coyle, 1984). Conceptual models are often used as intermediate steps towards simulation models (Robinson, 2008) based on SD. In the literature, there are cases where causal-loop diagrams and SD-based simulation models are used to conceptualise and formulate some challenges related to the energy access (Alam, 1997; Steel, 2008; Brent et al., 2011; Musango et al., 2011; Zhang & Cao, 2012; Jordan, 2013; Hartvigsson, 2015, 2016, 2018; Hartvigsson et al., 2015, 2018), as reviewed by the Author in (Riva & Colombo, 2020). In the latter, the Author formulates and calibrates the structure of a quantitative SD model based on a real case study in order to provide a robust modelling framework for analysing the electricity-development nexus.

From the review of the literature of the electricity-development nexus, the following main gaps can be identified:

- the main studies are conceptual analyses (Ahlborg & Hammar, 2014; Ahlborg & Sjöstedt, 2015; Riva, Ahlborg, et al., 2018) or they report upon observed rural electrification outcomes from real energy access projects (Wijayatunga & Attalage, 2005; Bastakoti, 2006; Peters et al., 2009; Kooijman-van Dijk & Clancy, 2010; Obeng & Evers, 2010);
- quantitative modelling studies do not focus on the analysis and identification of the main endogenous determinants of the nexus or the identification of enabling exogenous actions. In this regard, the Author investigated the existing approaches to model electricity demand in rural areas (e.g. trend analysis, scenarios, extrapolation, system-dynamics, agent-based modelling) (Riva, Tognollo, et al., 2018; Riva, Colombo, Piccardi, 2019), identifying that system-dynamics can represent an appropriate method since it can deal with the issue of data scarcity and provides a structural and holistic modelling framework for conceptualising and formulating the determinants and complexities affecting the evolution of electricity demand in unelectrified areas.

In order to fill this gap, building upon the modelling framework developed and validated by Riva et al. (Riva, 2019; Riva, Colombelli, et al., 2019; Riva & Colombo, 2020), the goal of this paper is to quantitatively assess the main fundamental dynamics, variables, and exogenous policies that characterise the electricity-development nexus, in order to derive useful guidelines for supporting future electrification actions in developing countries, with a particular focus on sub-Saharan Africa. The modelling framework is based on a real case-study, i.e. a hydroelectric-based electrification programme implemented in the rural community of Ikondo, Tanzania, in 2005 by the Italian NGO named CEFA Onlus. This can be perceived as a limitation of the approach, since the results achieved by following this approach lack of generalisation and immediate applicability to different contexts. On the other hand, in the relatively new and exploratory research field on energy access many issues are still unsolved, the availability of data is scarce, and the understanding and investigation of the electricity-development nexus and the evolution of electricity demand are still involving exploratory research approaches. This explains the decision of the Author to build their analysis upon the model developed by Riva et al. based on a real case study, which allowed grounding the understanding of the modelled issue on actual observed dynamics and quantitative data, paving the way for further simplifying and generalising the model.

The outline of this paper is as follows: **Material and methods** Section reports the description the case-study, an explanation of the model used for the analysis, and an introduction to the tests used to assess the model behaviour and outputs. Section **Results and Discussion** reports the results and a discussion of the tests, as well as the main policy implication in the field of rural electrification. Section **Conclusion** draws the conclusions and indication for future research work to overcome the limitations of the approach.

Material and methods

The Ikondo case-study

CEFA Onlus is an Italian NGO founded in 1972 by a group of agricultural cooperatives based in Bologna, Italy. In Tanzania, the NGO has realised three mini hydro-electric power plants in the rural areas of Iringa and Njombe regions, which currently serve around 10 villages connected to the mini-grids. The three plants are physically located in the villages of Matembwe, Bomalang'ombe, Ikondo. The village of Ikondo is the target context of this work. According to the latest 2016-data gathered by CEFA, Ikondo has a population of around 4000 people, divided in approximately 820 households (HHs). The village is characterised by an agriculture-based livelihood, with around 100–120 Income Generating Activities (IGAs) started after the electrification of the community in 2005, towards the installation of a run-of-river hydropower plant built in 2005 with a nominal power of 83 kW. In 2016, CEFA completed the upgrade of the power facility with an additional 350 kW turbine – achieving an overall generation capacity of 433 kW (Fig. 1). The power plant is managed by a local utility, namely Matembwe Village Company (MVC), founded by CEFA and now independent. CEFA founded also a local micro-credit utility, namely SACCOS, which is also in charge of recording and collecting the monthly electricity bills in the village.

The village was selected given the availability of historical measured data of electricity demand by the consumers connected to the distribution grid, starting from the first month of operation of the plant. Moreover, being considered a “successful case” of rural electrification by local people and practitioners, the selection of this case was driven by the possibility to derive modelling insights and useful guidelines from this rural electrification experience. The success of this project from a socio-economic perspective is discussed in the previous work of the Author (Riva & Colombo, 2020). To give two examples, according to local interviews, the number of Income Generating Activities (other than farming) is reported to have been increased from a couple of units



Fig. 1. Micro-hydro plants of Matembwe and Ikondo.

(before the project) up to more than 100 in 2018. Also, the manager director of the MVC indicated that respect to the initial situation, the average income of the village is more than tripled.

Data collection was the first implemented activity. Local visits in the field were carried out in three time-slots between 2016 and 2018 by the team of researchers of Politecnico di Milano, and the current Project Energy manager of CEFA – details on the type of data and uses are represented in [Riva and Colombo \(2020\)](#). Totally, 71 households (HHs) and 43 Income Generating Activities (IGAs) were interviewed in order to define ranges of income, expenditures, and working times, and to gather information about the electricity use (n° and type of appliances, operation time, time windows, power). 2 schools representatives, 1 hospital representative, 3 owners of an IGA, 2 farmers, 2 women, and 6 CEFA's representatives have been consulted and interviewed for understanding the history of development of Matembwe-Ikondo, conceptualising (i.e. identifying the main variables and dynamics) the electricity-development nexus for Ikondo, discussing the structure of the model, and collecting the data of monthly electricity consumption for Ikondo consumers from 2005 to 2017.

The system-dynamics model

Based on the analysis of the electricity-development nexus carried out by Riva et al. in [\(Riva, Ahlborg, et al., 2018\)](#), and through the on-field interviews conducted in Ikondo, the key variables and the main causal-loop diagrams for Ikondo were developed in [Riva and Colombo \(2020\)](#). In their work, the full and transparent formulation of the model is reported, that is the specification of its mathematical structure – i.e. the conversion of the feedback diagrams to algebraic, differential, and integral equations –, as well as the estimation of the parameters' value [\(Sterman, 2000\)](#). The simulation framework used was based on Vensim DSS® software. Its formulation followed an iterative process based on the questionnaires filled in during the on-field experience, the information shared with the local experts, and the analysis and redefinition of the structure by the modeller. The formulation of the final structure of the model is based on 11 main sub-models: *IGAs formation and Income, Market demand, Market production and revenues, Agricultural revenues, Population, Time savings, Education, IGAs electricity connections, HHs electricity connections, Household appliances diffusion, and Electrical Energy consumption*. It is composed by 80 integral equations, 260 auxiliary equations (i.e. algebraic and differential), and 140 constants. For each discrete time-step dt , the software solves all the equations that describe the dynamic behaviour of the modelled system. For formulating individuals' behaviour, to deal with the classical SD-hypothesis of perfect mixing and homogeneity within the levels, the agents of the model are represented as households divided in two classes: low income households (LI HHs) (i.e. who rely only on agricultural-based activities) and high-income households (HI HHs) (i.e. who rely both on agricultural-based activities and work in an IGA).

The model simulates the impact of electricity access and its use on the socio-economic development experienced in Ikondo, and the related feedbacks on the community's electricity consumption. One of the outputs is therefore the projection of long-term electricity demand of all the local households (both LI and HI HHs) and IGAs (i.e. the loads of the facilities used by people to run their income activity) along the simulated time horizon.

Model testing

"A model is a simplification, an abstraction, a selection, because our models are inevitably incomplete, incorrect – wrong" [\(Sterman, 2002 pg. 525\)](#). These are the words used by John Sterman in his very famous Jay Wright Forrester Prize Lecture in 2002 for discussing the complexity of model testing within the systems science. In the past, this consideration that models are obvious simplifications of the real world allowed the social science community to overcome the classical *logical positivism*

theory of knowledge, which asserted that a statement is cognitively meaningful only if it is verifiable through empirical observation [\(Barlas & Carpenter, 1990\)](#). System thinking theory supports the opposite thesis: testing a model just for "proving" that it is "right" – e.g. by focusing just on the statistical fitting of historical data – limits the utility and credibility of a model. Indeed, "[...] because all models are wrong, we reject the notion that models can be validated in the dictionary definition sense of 'establishing truthfulness', instead focusing on creating models that are useful, on the process of testing, on the ongoing comparison of the model against all data of all types, and on the continual iteration between experiments with the virtual world of the model and experiments in the real world" [\(Sterman, 2002 pg. 521, 525\)](#)".

In accordance with Sterman's words, the testing of the model developed by the Author aims at deriving elements of usefulness and policy guidelines in the field of rural electrification based on the Ikondo case. 3 types of SD specific tests are carried out: (1) *Behaviour sensitivity* test, (2) *Family-member* test, (3) *Policies* test. In addition, to further assess the model structure and its policy implications, the analysis is extended to evaluate the relevance of two some socio-economic dynamics of the model, viz. the nexus between electricity and (1) time availability, (2) local market improvements. This contributes to assess further dynamics of the electricity-development nexus that are debated in the literature [\(Riva, Ahlborg, et al., 2018\)](#).

Results and Discussion

Behaviour sensitivity test

The behaviour sensitivity test is useful for determining those parameters to which the model is highly sensitive [\(Barlas, 1996\)](#). It contributes to build confidence in the model by checking that also the real systems would exhibit similar high sensitivity to the corresponding parameters. Moreover, it helps identifying the most important and critical aspects that affect the rural electricity-development nexus from a quantitative point of view.

The test was conducted empirically, by evaluating the sensitiveness of the calibration results of the model to the univariate variation of each calibrated parameter (totally, there are 123 parameters in the model [\(Riva & Colombo, 2020\)](#), as reported also in the Supplementary Material). The indicator used for testing the sensitiveness was the calibration payoff, which is the number that summarizes the goodness of the model output respect to the available data. In the model, the payoff was calculated according to an iterative method that starts by simulating the model with an initial set of values for the calibrating parameters. At each iteration k performed by the optimiser, the weighted difference between the historical datum and the simulated value of the corresponding model variable is computed for each time t . The opposite of the sum of all such differences computed along the entire horizon T gives the payoff of the simulation k . The final iteration performed by the optimiser is the one with the minimum value of the payoff that determines the end of the calibration.

Given the payoff, the *Behaviour sensitivity* test is carried out as follows: for each one of the 123 parameters at time, the variation necessary to cause a change of 20%, 15%, 10% and 5% in the model payoff is calculated. The numbers of parameters causing those changes in the payoff through variations lower than the 20%, 15%, 10%, and 5% thresholds are tracked in [Fig. 2](#).

The number of critical parameters gradually decreases when taking into account higher payoff changes, viz. the higher is the desired payoff change, the lower is the number of parameters that, one at a time, generates such payoff change. Also, given a certain payoff change, the number of critical parameters decreases by reducing the variation threshold, viz. the lower is the variation of the parameters, the lower will be their impact on the payoff change. The most critical parameters are the ones on the right-side of the figure in the dashed circle, viz. the parameters that cause a change of 20%, 15%, 10%, and 5% on the payoff by varying

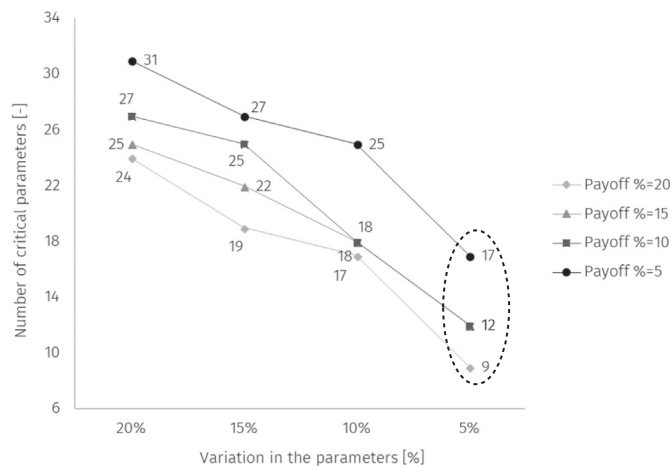


Fig. 2. Number of critical parameters on varying the changes tested on the payoff.

less than 5%. For checking that the real system would exhibit similar high sensitivity to the corresponding parameters, in the next section discusses the analysis of the two extremes, viz. the 17 and 9 parameters that cause a change of 5% and 20% on the payoff, respectively.

Family-member test

According to Sterman (2002b), “the family member test asks whether the model can generate the behaviour of other instances in the same class as the system the model was built to mimic”. We relied on this test for discussing the results of the model when its inputs take on different values, until the extreme ones. The results provide useful insights about complementary activities needed to support a positive rural electricity-development nexus.

This test was performed by using the 17 most sensible parameters found through the *Behaviour sensitivity* test as main indicators. The test aimed at evaluating the response of the model by varying such parameters *ceteris paribus* within a feasible physical interval and then analysing the simulation results on some of the most meaningful variables. It was verified that the model response is in line with the expectations.¹ When the lower/upper bounds could not be physically defined, the model was stressed to the lowest/highest possible value (indicated as “tested” in Table 1), until the occurrence of dynamics not considered in the model boundaries (e.g. no more only-agricultural reliant LI HHs). In Table 1, the model behaviour is discussed by displaying some of the most meaningful variables of the model by varying the related sensible parameter by -20% , -10% , $+10\%$, $+20\%$ respect to the calibrated value, and reporting useful insights for each. The full explanation of the model variables reported in the table below can be found in Riva and Colombo (2020).

Policy test

This test aims at exploring model behaviour when subjected to different policies and exogenous decision-making processes. The final goal is to perform a kind of *what-if* analysis and derive some useful insights on the policies implemented by CEFA, potential improvements, and useful lesson learnt that can be extended to the rural electrification field. In some cases, performing such tests required some changes in the model structure, which are described further.

Micro-credit

The effect of micro-credit is here assessed by exploring the following two scenarios:

- absence of micro-credit;
- total access to microcredit (all people used micro-credit to set-up a business).

The results displayed in Fig. 3 indicate that access to micro-credit had a significant impact on the socio-economic development of the local community and the consumption of electricity. In case of no access to micro-credit, the number of businesses, the average income of the village, and the number of people that get access to electricity is reduced by half, since people have fewer financial resources to use. All these effects cause a significant lower total electricity demand. On the other hand, these outputs suggest also that a partial access to micro-credit, as in the calibrated case-study, does not lead to significantly different results respect to the case of complete access to micro-credit. Rather, looking at the first two figures on the top-side of Fig. 3, it seems that the case of complete access to micro-credit leads to an initial phase of oscillations in the number of IGAs. This can be explained by considering that with more access to micro-credit, more people can afford the setting-up of a new business, speeding-up the process of financing, and causing a continuous overshooting and undershooting dynamics that slowly gets damper and then stabilises.² These oscillations suggest that access to micro-credit in rural for setting-up productive activities should be always combined with carefully demand-supply market dynamics, in order to avoid overinvestments in activity and generate unsustainable situation of market competition.

Cost of electricity and social responsibility

Almost all the interviewed people in Ikondo-Matembwe indicated the electricity cost set by the local utility MVC to be very affordable. The effect of the electricity price is here assessed by exploring the following two scenarios:

- increasing the variable electricity cost by 5 after 1 year, by keeping constant the fraction of revenues that the MVC reinvests within the local community (viz. with social responsibility case);
- increasing the variable electricity cost by 5 after 1 year, by setting the fraction of revenues that the MVC reinvests within the local community equal to 0 (viz. without social responsibility case).

The outputs in Fig. 4 confirm that increasing the electricity cost, obviously, decreases the financial availability of HHs for goods and services, especially the LI HHs. In case of social responsibility from the utility, this does not impact on the socio-economic development of the village, while it causes a decrease in the total community electricity demand, especially for the EE-reliant IGAs. Indeed, according to the model output, the electricity expenditures with the current electricity tariff implemented by the MVC represent around 10% of their profit. Instead, HHs are used to pay a lot for traditional sources of energy (e.g. kerosene, candles, paraffin), and their electricity demand elasticity is indeed very low. Moreover, the expenditures for electricity with the current electricity tariffs implemented by the MVC represent around 1–2% of their total income, that is a very small fraction. In the case of absence of social responsibility, increasing the electricity cost impacts also on the number of IGAs and the average income, due to a decrease in the HHs market demand. These results suggest that increasing the cost of electricity does not alter the socio-economic structure of local communities if a reinvestment of the utility revenues within the local community is

¹ Where not otherwise specified, the consideration of the model's response (e.g. “higher”/“lower”) is respect to the calibrated simulation.

² In the model (Riva & Colombo, 2020), this dynamics is caused by the fact that the decision of setting-up a new IGA is based just on the actual existing number of IGAs and the perceived needed ones, neglecting the work-in-progress IGAs in the supply chain (i.e. IGAs to be financed, and IGAs already financed to be set-up).

guaranteed. This is an interesting result, especially useful when exogenous or endogenous events require an increase of the electricity tariff. On the other hand, any increment is followed by a significant decrease of electricity use, which must be considered in the planning phase of the electricity system, in order to avoid shocks in the tariff values, especially for the EE-reliant businesses.

Another notable result is the different sharp drops of electricity demand between the total residential electricity demand and the productive monthly electricity demand. These drops, explained in Riva and Colombo (2020), are the effect of the change in tariff from one year to the next. The figures indicate that the productive use of electricity is much more sensitive to step-changes of electricity price respect to the residential one.

Electricity reliability

Electricity reliability is one of the most important factors to consider for implementing sustainable electricity plans in rural communities, as emerged by the literature on the electricity-development nexus (Riva, Ahlborg, et al., 2018), and stated by Hartvigsson et al. (Hartvigsson et al., 2015, 2018; Hartvigsson, 2018).

In the case of Ikondo, the local surveys and data confirmed the high quality of the electricity service delivered to people. For this reason, the reliability parameter was introduced in the model as an exogenous measure of “unavailability” of the service, which has a feedback just on the actual time of electricity use (Riva, 2019; Riva, Colombelli, et al., 2019). In order to consider the impact of electricity reliability on people willingness to have access to and consume electricity, the

Table 1

Tests on the most sensible parameters.

Parameter	Extreme conditions and graphical insights	
Initial external market demand / fr change in external market demand / Initial external agricultural expenditures	MIN: 0 MEANING: simulation of a closed economy RESPONSE: average income of the village remains lower; less connections and so electricity use.	MAX: tested MEANING: simulation of a market with high trading level. RESPONSE: higher economic progress; higher number of IGAs; average income of the village increases; higher number of connections and so electricity demand.

These results confirm that the positive impacts of electrification on a rural context are less evident when the local economy is more closed to trade (due to, for example, geographical or adverse socio-economic conditions). This indicates that when electrification takes place in rural contexts where trading level is high, local people can build upon this favourable trading condition to exploit at a higher degree the benefits of having electricity available. This may suggest that supporting electrification with complementary activities for improving the accessibility, and so the potential trading level, of isolated villages may reveal as a useful strategy. This conclusion is line with Lenz et al. (Lenz et al., 2017), who report that in rural Rwanda, only rural communities located next to a main road and frequented by casual customers from outside experienced a net increase in income through sales of improved services and goods.

Another interesting outcome of this test is that the variable representing electricity connections (“Total connected HHs”) show less variance respect to the variable (“Connected IGAs”). This confirmed also what emerged from local interview with local people and experts, who indicated that electricity access is perceived as a fundamental good by locals, who tend to save money to invest in having electricity at home. This necessity to save money is related to the business model adopted in Ikondo: households have to pay for the connection and, although they perceive access to electricity in their house as a pivotal and fundamental asset, they need to accumulate money for making the investment. This is the reason why the electrification path in Ikondo took about 11 years to connect 150 households. On the contrary, the number of IGAs connected is more dependent on the variability of the parameter representing the market turnover, since less market demand causes a smaller number of IGAs.

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Table 1 (continued)

Parameter	Extreme conditions and graphical insights	
fr change in external food expenditures	MIN: 0 MEANING: simulation of a closed agricultural economy RESPONSE: income of the LI HHs remains much lower; income of HI HHs decreases very little; HHs and IGAs connections reduce by causing a large of reduction of electricity demand.	MAX: tested MEANING: simulation of a market with high trading level. RESPONSE: high economic progress reflected in the average income of the village that increases for both the LI HHs and the HI HHs. Higher number of connections and electricity demand.

HI Income

LI Income

Total monthly electricity demand

These results confirm that changes in the agricultural livelihood and dynamics largely impact on the electrification output (i.e. the variable “Total monthly electricity demand” largely varies – the sharp drops along the trend is the effect of step-changes of electricity tariff, as explained in (Riva & Colombo, 2020) and also reported in sub-section “Cost of electricity and social responsibility” in section Policy test. This is especially due to the fact that the income of LI HHs (variable “LI Income”), who are farmers, is highly dependent on the demand, and so the selling of agricultural products. On the contrary, the variable “HI Income” shows less variable dynamics, and this is due to the fact that HI HHs have also another source of income generated by their IGA other than the selling of agricultural products, which is less sensitive to variations of the tested parameter. From the HI Income and LI Income, it emerges an initial phase of sudden decrease of the income, which is due to the initial high level of unsatisfied market demand, which creates a backlog of perceived needed new IGAs. To set-up all such initially desired IGAs, the initial level of income invested to set-up the business is high. This decrease of HI Income variable takes almost 1 year (i.e. 50 weeks) to recover to the initial level, suggesting that when electricity access does open up to new business opportunities, it takes time to reveal its economic benefits to local people.

These results suggest that agricultural activities play an important role also after electrification and the creation of new business. For example, CEFA experts stated that in the village of Bomalang’ombe, close to Ikondo, many farmers abandoned agricultural activities after the creation of new IGAs, limiting the overall benefits of electrification on the local community, creating disparities and too much dependence on electricity quality and reliability. Moreover, this indicates that implementing awareness campaigns intended to improve farming productivity for attracting external consumers is definitively a win-win strategy.

model has been modified by adding a new variable, namely “customer’s response” (CR) to electricity quality. According to Hartvigsson et al. (2018), the variable can be expressed by a logistic function, which depends on the reliability of electricity:

$$CR = \frac{1}{1 + e^{-k_{cr} \cdot (R - R_0)}} \quad (1)$$

The parameter k_{cr} influences the steepness of the logistic function (Fig. 5, left-side) – viz. the higher it is, the more unexpected would be the response of the customer around a certain crucial value of reliability. The parameter R is the reliability (i.e. unavailability) of the electrical supply, and R_0 is the crucial value of the R called “sigmoid’s

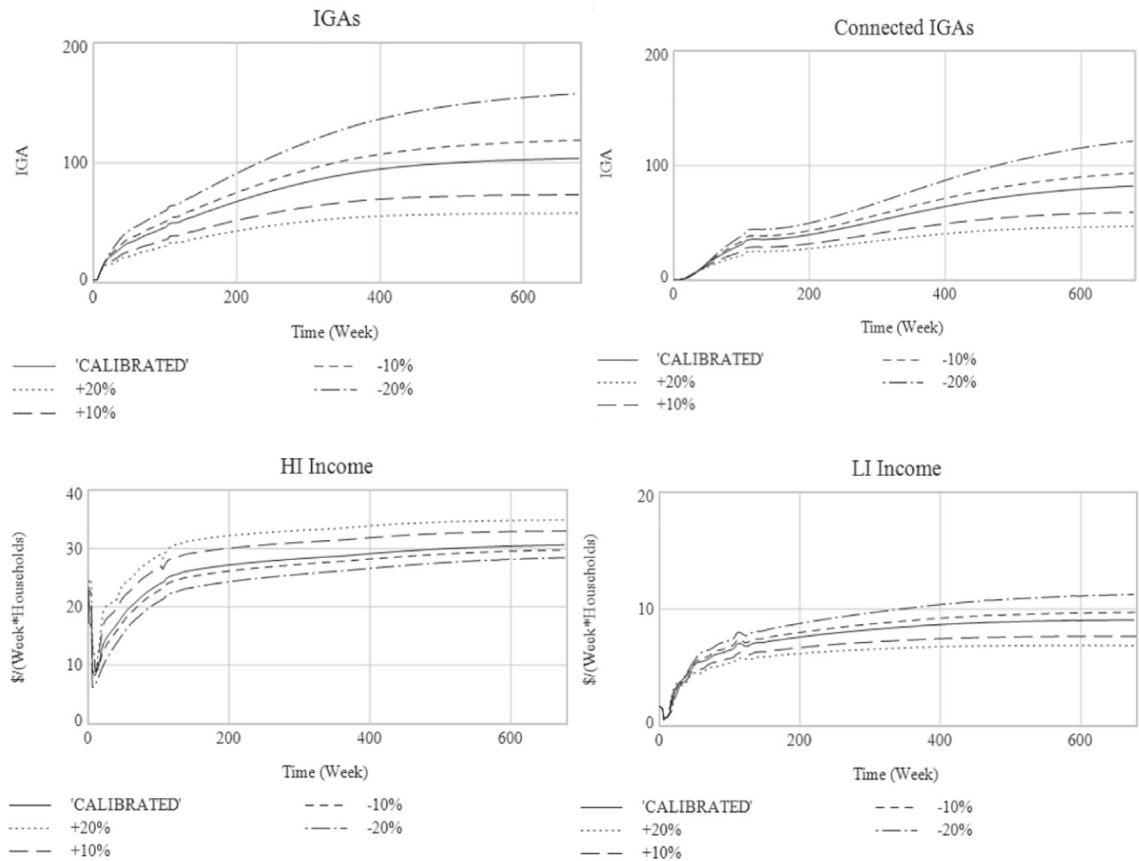
midpoint” – viz. where the variable *customer’s response* is reduced to 50% of its maximum (Fig. 5, right-side).

To test the effect of customer’s response to changes in the power reliability, the model behaviour is explored by decreasing the variable representing the electrical reliability by 0.01%, 0.05%, and 0.1% per time step. The parameters k_{cr} and R_0 are set equal to 25 and 0.31 in accordance with the mean values taken from (Hartvigsson et al., 2018).

As expected, decreasing the quality of the electricity service, electricity connections and use are lower respect to the calibrated case. In some cases, the consumption even starts decreasing after a certain point (Fig. 6). The assumption here is that people do not disconnect, even for low levels of reliability, since the fixed component of the tariff to pay every month is very low; otherwise, the impact would be even

Table 1 (continued)

Parameter	Extreme conditions and graphical insights	
Reference factor productivity	MIN: tested MEANING: simulation of a market with very high competition. Local IGAs are able to satisfy a very little fraction of the market demand. RESPONSE: very high number of IGAs formation; little decrease in the HI income, due to market competition; little increase of LI income, since there are more rich people that spend more in food; sensible increase in the connected IGAs and IGAs electricity demand.	MAX: tested MEANING: simulation of a market with very little competition, since the few existing IGAs are able to satisfy alone the market demand. RESPONSE: very few formation of new IGAs; HI income of the few IGAs is higher, due to the absence of market competition; little decrease of LI income, since the new households caused by the increase in the population are all farmers, enforcing the vicious circle of poverty.



These results suggest that in contexts characterised by an already high level and quality of labour, electricity use benefits less people (lower "LI Income") and businesses (lower "Connected IGAs", which show a change in the simulated trend at around week 100 due to a change in the connection fee implemented by CEFA), increasing inequality. To avoid this, these results suggest that providing access to electricity in such contexts should be supported by complementary activities aimed at innovating the local production, increasing the market competition on new goods and services. This is also supported in the literature: In fact, one of the most recurrently identified obstacles to the expansion of rural business is the lack of a dynamic local market (Neelsen & Peters, 2011; Kooijman-van Dijk, 2012; Baldwin et al., 2015), leading to the "crowding out effect" of the existing firms, i.e. the creation of new IGAs that is followed by stagnation or economic losses among already existing IGAs (Kooijman-van Dijk & Clancy, 2010; Peters et al., 2011), or a reduction of wages due to an abundance of labour supply over labour demand (Dinkelman, 2011). Moreover, this result points out a potential limit of the supply-demand model due to the perfect mixing hypothesis of SD. In this formulation, there is no diversification between businesses, since all of them satisfy the market demand equally and operate under "perfect competition" rules. A further disaggregation based on people's ranking of priorities – e.g. essential goods vs luxury goods – can be a solution for achieving more realistic results, but the larger is the disaggregation, the more appropriate would be the transition towards Agent-based models (ABMs)

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worse, causing a high rate of disconnections, as demonstrated also by Hartvigsson et al. (2018). This result confirms the need to introduce the concept of reliability and/or consumer satisfaction in the traditional "binary" metrics – viz. access or not access to electricity – usually employed for tracking the global progress in providing universal electricity access. Indeed, using just the initial number of connections as an indicator for electrification programmes can cause low reliability and unaffordable electricity prices, due to the underestimation of the affordability- and reliability-related complexities.

Testing further socio-economic dynamics

These tests aim at evaluating the importance and the impact of electricity access on some socio-economic dynamics, and the reverse feedback.

Electricity-use of time

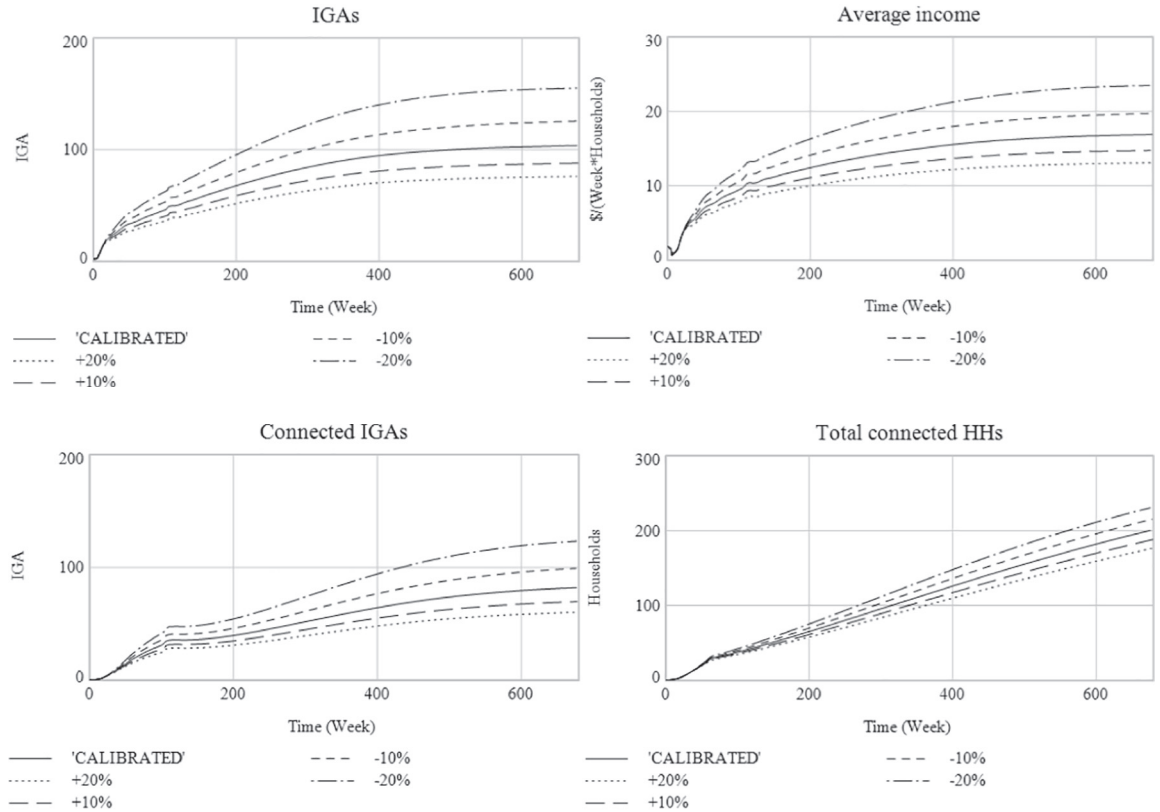
As emerged from the survey carried on in Ikondo-Matembwe, almost all the interviewed people mentioned the time available during

Table 1 (continued)

Parameter	Extreme conditions and graphical insights	
Fraction of feasible HHs market supply	MIN: tested MEANING: simulation of a market able to satisfy 0% of the local HHs demand. RESPONSE: lower number of IGAs formation, which satisfies just the supply-chain of other IGAs and the external demand. HI income remains unchanged respect to the initial condition; lower total electricity demand.	MAX: 1 MEANING: simulation of a market able to satisfy 100% of the local HHs demand. RESPONSE: higher number of IGAs formation; little increase in the HI income, due to a higher market demand but also a higher market competition; higher total electricity demand.
	Total market demand IGAs HI Income Total monthly electricity demand These results highlight that a local context which could potentially have the resources for satisfying all the local market demand is very suitable for introducing access to electricity. On the contrary, a very isolated village with no resources would not develop just with access to electrification, but it needs complementary activities for obtaining enough productivity inputs and resources. This is suggested also by Kirubi et al., who indicate that to facilitate such a development after access to electricity, other infrastructures such as roads and telecommunications need improvements, as these can reduce transactions costs and make rural IGAs “competitive in out-sourcing of business services and products destined for the lucrative urban markets” ((Kirubi et al., 2009) p. 1219).	
Initial Population	MIN: 1/3 of the calibrated value MEANING: simulation of a smaller community. RESPONSE: very low electricity demand. This is due to the fact that the number of IGAs formation is lower because of the backlog effect of the initial “perceived” needed new IGAs. This backlog causes the set-up of too many IGAs, which create a too high delayed market competition and then a delayed decrease in HI Income. In turn, this causes a delayed decrease in the HI Income.	MAX: equal to the carrying capacity MEANING: simulation of a larger community. RESPONSE: higher electricity demand. This is due to the fact that the number of IGAs formation is higher due to initial higher unsatisfied market demand.
	Total residential monthly electricity demand Total productive monthly electricity demand These results suggest that, ceteris paribus, more populated villages can gain more from access to electricity. This could have an important policy implication: in electrified areas, electrification programmes can start from bigger communities in order to foster spillover effects on other smaller communities. The two graphs also show that the extent of the sharp drops along the trends of the two simulated variables due to the step-changes of electricity tariff is different. This indicates that the elasticity of electricity demand is much higher for businesses – i.e. electricity consumption for productive purposes is more sensitive to change in the tariff, especially due to the fact the some business as mills are electricity-intensive IGAs.	

Table 1 (continued)

Parameter	Extreme conditions and graphical insights	
Initial number of IGAs	MIN: tested (~1) MEANING: simulation of almost absence of initial IGAs RESPONSE: higher number of IGAs formation due to initial higher unsatisfied market demand; increase in both HI income and LI income, due to a higher internal and external market demand; increase in both connected IGAs /HHs and so of IGAs/HHs electricity demand.	MAX: tested MEANING: simulation of initial IGAs able to satisfy the local demand. RESPONSE: very low IGAs formation due to population increase. HI and LI average incomes increase less. Lower number of connected IGAs /HHs and so lower IGAs/HHs electricity demand.



These results suggest that contexts characterised by an already high number of IGAs, *ceteris paribus*, experience less proportional benefits from electrification (less increase of “Average Income”), especially because the trend of market development is smoother, attracting less external market demand. The results explain also the so-called “economic boom” of Ikondo-Matembwe villages, where people were all subsistence farmers before electrification. Moreover, as for *Reference market productivity*, this result points out a potential limit of the supply-demand model due to the perfect mixing hypothesis of SD. In this formulation, there is no diversification between businesses, since all of them satisfy the market demand equally and operate under “perfect competition” rules. A further disaggregation based on people’s ranking of priorities – e.g. essential goods vs luxury goods.

(continued on next page)

the evening hours as an important effect of electricity use. The simulations (Fig. 7) indicate that the evening hours are employed especially for productive purposes, rather than farming. Moreover, the trend of the *operation time* curve starts decreasing at a certain point. This is caused by the change in the electricity connection tariff in 2006, which created a fractional decrease in the IGAs connection rate, without affecting the rate of IGAs formation. The trend keeps increasing again as the number of IGAs starts reaching the plateau.

The feedback on the economic dynamics and the electricity demand is displayed in Fig. 8. Removing the feedback of the time-variable has a non-negligible impact on the number of IGAs and HI Income. In particular, the number of IGAs increases, since the local market demand cannot be satisfied in the evening and more IGAs are needed for increasing the daily production. This is due to the modelling hypothesis that the number of new IGAs increases based on the gap between demand and supply: with no time for working in the evening, the supply of local demand during only day hours requires more enterprises. Nevertheless, this is in line with

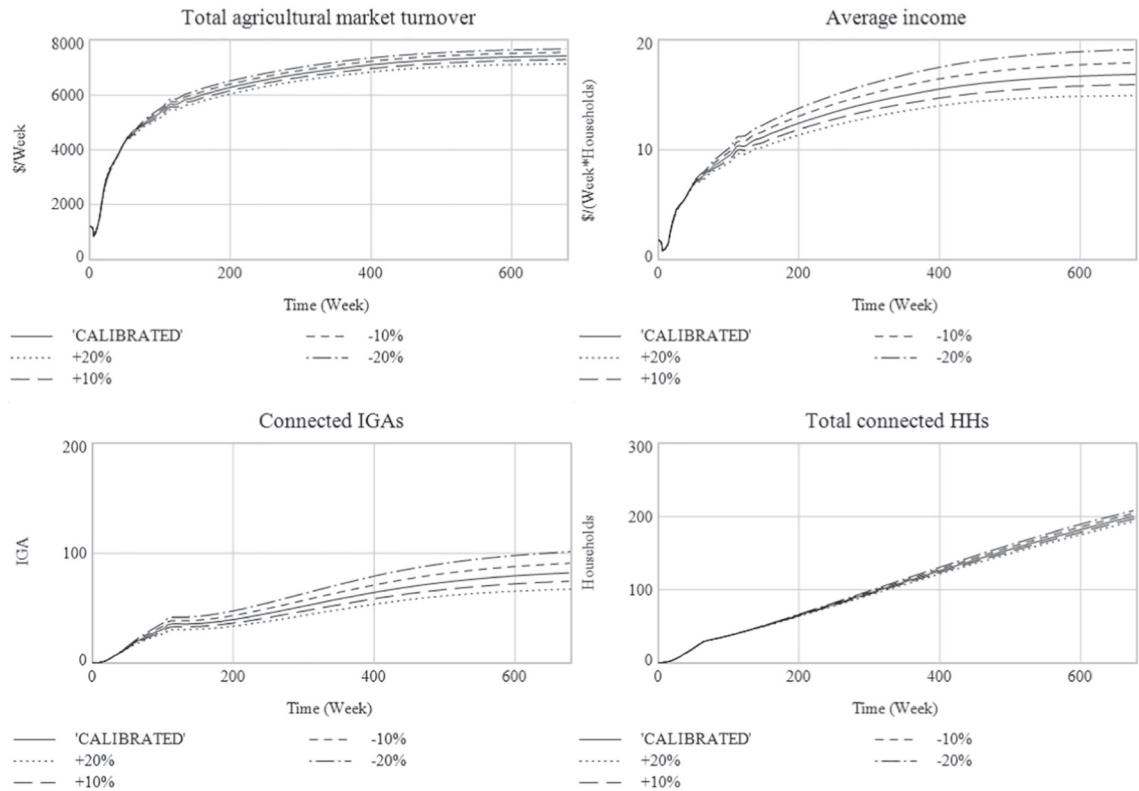
the literature which does not report an increase of businesses as a result of night lighting. On the contrary, the decrease in the HI Income is in line with the local surveys, which suggested an increase in the sales and revenues as a result of the evening working. Fig. 8 confirms also that from an aggregated point of view, the feedback of time freed-up by electricity use on the community demand is negligible (i.e. demand 3% higher at the end of the time horizon). On the other hand, focusing just on the IGAs demand, the disaggregated feedback generates an increase of 11% in the electricity use, since there are more IGAs.

Electricity-market improvements

This test aims at assessing the effect of electricity use on the local market demand for new local products with higher quality, and the feedback on electricity use. The same behaviour is generated by investigating the effect of price reduction due to market expansion because of electrification. The results displayed in Fig. 9 show that neglecting these dynamics would significantly bias the estimation

Table 1 (continued)

Parameter	Extreme conditions and graphical insights	
Initial fr income for food expenditures HI	MIN: tested MEANING: less HI expenditures for food and more for market. RESPONSE: increase in the average income since HI income, due to higher market demand (HI HHs spends more for market goods and services), increases, and so does LI income: although the fraction of HI expenditures for food is lower, since the HI income is higher and there are more HI HHs, the HI expenditures for food are higher in the absolute term, albeit slightly; increase in the connected IGAs; lower proportional increase in the number of connected HHs.	MAX: → 1 MEANING: HI HHs spend almost all their income in food RESPONSE: lower number of IGAs formation and connection, which satisfy just the supply-chain of other IGAs, the external demand, and the LI demand; average income slightly lower than the same level of the calibrated simulation, since for HI Income the lower demand is balanced by a lower number of IGAs, while for LI income there are more fraction of HI income spent for food, but with lower HI Income; little decrease in HH connections and so lower HHs electricity demand.



These results indicate that in contexts where people live just on subsistence farming, electricity use alone cannot boost sustainable development. In such contexts, electrification programmes should therefore be supported by activities and awareness campaign aimed at diversifying people's basket of goods and services. In this context, the diffusion of communication devices – e.g. TVs, radio and phones – may help, since they can introduce changes in aspirations and expenditures of rural households (Matinga & Annegam, 2013) for goods and services, diversifying purchases and leading people to shop locally rather than elsewhere (Shackleton et al., 2009).

The first two figures on “Total agricultural market turnover” and “Average income” indicate also that the largest “boost” to the local economy takes place in the initial years – by the first 200 weeks (i.e. around 4 years), the largest increase in the two variables can be observed. This suggests that when developing access to electricity projects, the first years after their implementation are the most critical in terms of monitoring, since they are the years when the largest development dynamics and structural changes take place, as well as the associated potential risks.

of the final community electricity demand. This confirms again an important lesson to consider when measuring the impact of electrification actions: the level of isolation of a village and the proximity to other electrified areas are fundamental conditions to take into account. In villages well connected with other electrified areas, the relative impact of electrification would be considerably lower, since such villages are benefited from a spill-over of the surrounding areas. This is evident also from the surveys in the unelectrified village of Kitole, close to Matembwe and Ikondo, who is characterised by a higher welfare level respect to Ikondo before electrification.

Conclusion

Around the world, around 770 million people do not have sustainable access to electricity. This is considered a limiting factor to the

socio-economic development of rural communities in developing countries. During the last decades, international donors, organisations, NGOs, universities, energy planners, practitioners, and private companies have been investing a lot of resources in programmes and projects that aim at improving rural people's socio-economic conditions through access to energy. Despite these investments, the scientific literature reports only fragmentary and sometimes controversial results regarding the impacts of energy programmes, and most of the times the causes are attributable to the inappropriate planning of the off-grid power system capacities. In this context, wrong projections and assessments of the long-term electricity demand contribute to the over- and under-sizing of electricity systems and related infrastructures, especially in off-grid areas. This is caused by a limited knowledge and modelling of the long-term impact of electricity access on local socio-economic development and the consequent feedback on electricity demand. Moreover, the scientific

Table 1 (continued)

Parameter	Extreme conditions and graphical insights	
Initial available operation time	MIN: tested MEANING: initial low time spent for working RESPONSE: higher number of IGAs formation (due to lower production per IGA); little decrease in the HI income, due to market competition; little increase of LI income, since there are more rich people that spend more in food; increase of total electricity demand as result of more IGAs that get connected and consume electricity.	MAX: 168 (viz. 24x7) MEANING: HI HHs spend all their time for working RESPONSE: formation of very few IGAs; HI income of the very few IGAs is much higher, due to the absence of market competition; LI income does not increase, since the new households caused by the increase in the population are all farmers, enforcing the vicious circle of poverty; large reduction of electricity connections and demand.
	The figure consists of four line graphs arranged in a 2x2 grid, showing the impact of initial available operation time on various socio-economic indicators over 600 weeks. Each graph compares a 'CALIBRATED' baseline (solid line) with +10% (dashed line), -10% (dotted line), and -20% (dash-dot line) variations. IGAs: The y-axis represents the number of IGAs (0 to 200). In the MIN scenario, IGAs increase over time, with the -20% variation reaching the highest value (approx. 180) and the +10% variation reaching the lowest (approx. 100). In the MAX scenario, IGAs decrease over time, with the -20% variation reaching the lowest value (approx. 50) and the +10% variation reaching the highest (approx. 150). HI Income: The y-axis represents HI income in \$/(Week*Households) (0 to 40). In the MIN scenario, HI income increases over time, with the -20% variation reaching the highest value (approx. 35) and the +10% variation reaching the lowest (approx. 25). In the MAX scenario, HI income increases significantly, with the -20% variation reaching the highest value (approx. 35) and the +10% variation reaching the lowest (approx. 25). LI Income: The y-axis represents LI income in \$/(Week*Households) (0 to 20). In the MIN scenario, LI income increases over time, with the -20% variation reaching the highest value (approx. 15) and the +10% variation reaching the lowest (approx. 5). In the MAX scenario, LI income remains relatively stable, with the -20% variation reaching the highest value (approx. 15) and the +10% variation reaching the lowest (approx. 5). Total monthly electricity demand: The y-axis represents electricity demand in Hour*kWatt/Month (0 to 20000). In the MIN scenario, electricity demand increases over time, with the -20% variation reaching the highest value (approx. 15000) and the +10% variation reaching the lowest (approx. 5000). In the MAX scenario, electricity demand decreases over time, with the -20% variation reaching the highest value (approx. 15000) and the +10% variation reaching the lowest (approx. 5000).	
fr change in internal IGAs supply / max fr of internal IGAs supply	MIN: 0 MEANING: the supply chain of IGAs is all outside the village RESPONSE: the total market demand decreases, with a consequent lower IGAs formation, and so lower growth of HI and LI Income; lower electricity demand.	MAX: tested MEANING: the supply chain of IGAs is all local RESPONSE: very high total market demand, which causes a higher number of IGAs, with a consequent increase of the average income of the village; higher electricity demand.
	The figure consists of two line graphs arranged in a 1x2 grid, showing the impact of fr change in internal IGAs supply on market demand and electricity demand over 600 weeks. Each graph compares a 'CALIBRATED' baseline (solid line) with +10% (dashed line), -10% (dotted line), and -20% (dash-dot line) variations. Total market demand: The y-axis represents market demand in goods/Week (0 to 4000). In the MIN scenario, market demand decreases over time, with the -20% variation reaching the lowest value (approx. 2000) and the +10% variation reaching the highest (approx. 3500). In the MAX scenario, market demand increases over time, with the -20% variation reaching the lowest value (approx. 2000) and the +10% variation reaching the highest (approx. 3500). Total monthly electricity demand: The y-axis represents electricity demand in Hour*kWatt/Month (0 to 20000). In the MIN scenario, electricity demand decreases over time, with the -20% variation reaching the lowest value (approx. 5000) and the +10% variation reaching the highest (approx. 15000). In the MAX scenario, electricity demand increases over time, with the -20% variation reaching the lowest value (approx. 5000) and the +10% variation reaching the highest (approx. 15000).	
This test provides useful insights concerning contexts characterised by very high inequality and disparity between different socio-economic classes. Where the entire demand is satisfied by very few market suppliers, electricity use has a very low impact on local development. Therefore, electrification programmes must first try to reduce inequality by socio-economic complementary initiatives. A further disaggregation of IGAs based on people's ranking of priorities – e.g. essential goods vs luxury goods – and removing the “perfect competition” market rules would probably reduce the impact of this parameter on the payoff.		
This output confirms that the diversification of local IGAs is an important determinant of local development. In contexts characterised by very low skills, electricity does not create expected benefits. Electricity use should therefore not just support an increase of the quantity of local business, but also the type and the quality. This suggests that electrification must be supported by effort devoted to increase people's skills in delivering new services and products, as confirmed also in the literature (Bastakoti, 2006; Kooijman-van Dijk, 2012).		

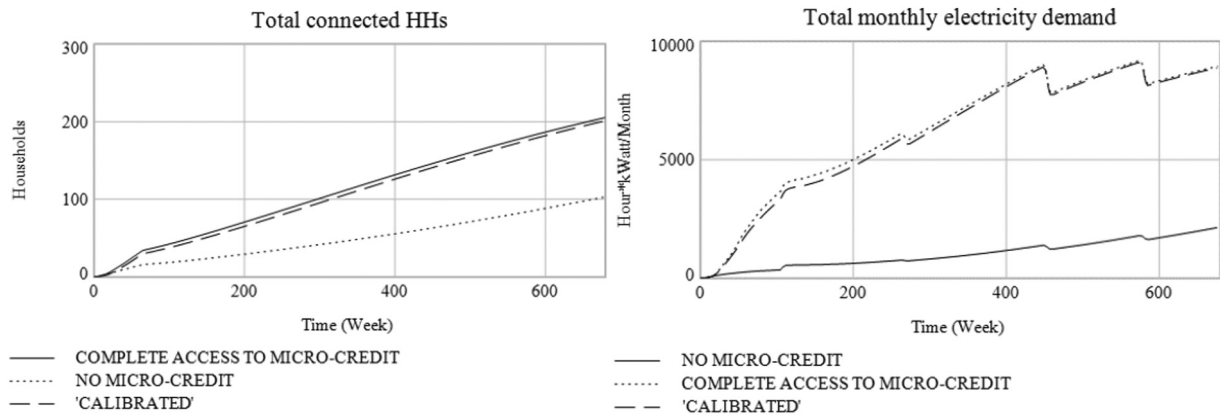


Fig. 3. Effect of micro-credit on some of the main outputs.

investigation of this topic is rarely coupled with the research on off-grid rural electricity planning, leading to an insufficient attention to “long-term aspects” in the traditional practices, models and methods for rural electrification.

In this paper, a system dynamics-based approach is employed to explore the electricity-development nexus experienced in a rural village of Tanzania, namely Ikondo. A simulation model developed by the Author is tested in order to derive insights about its outputs under different

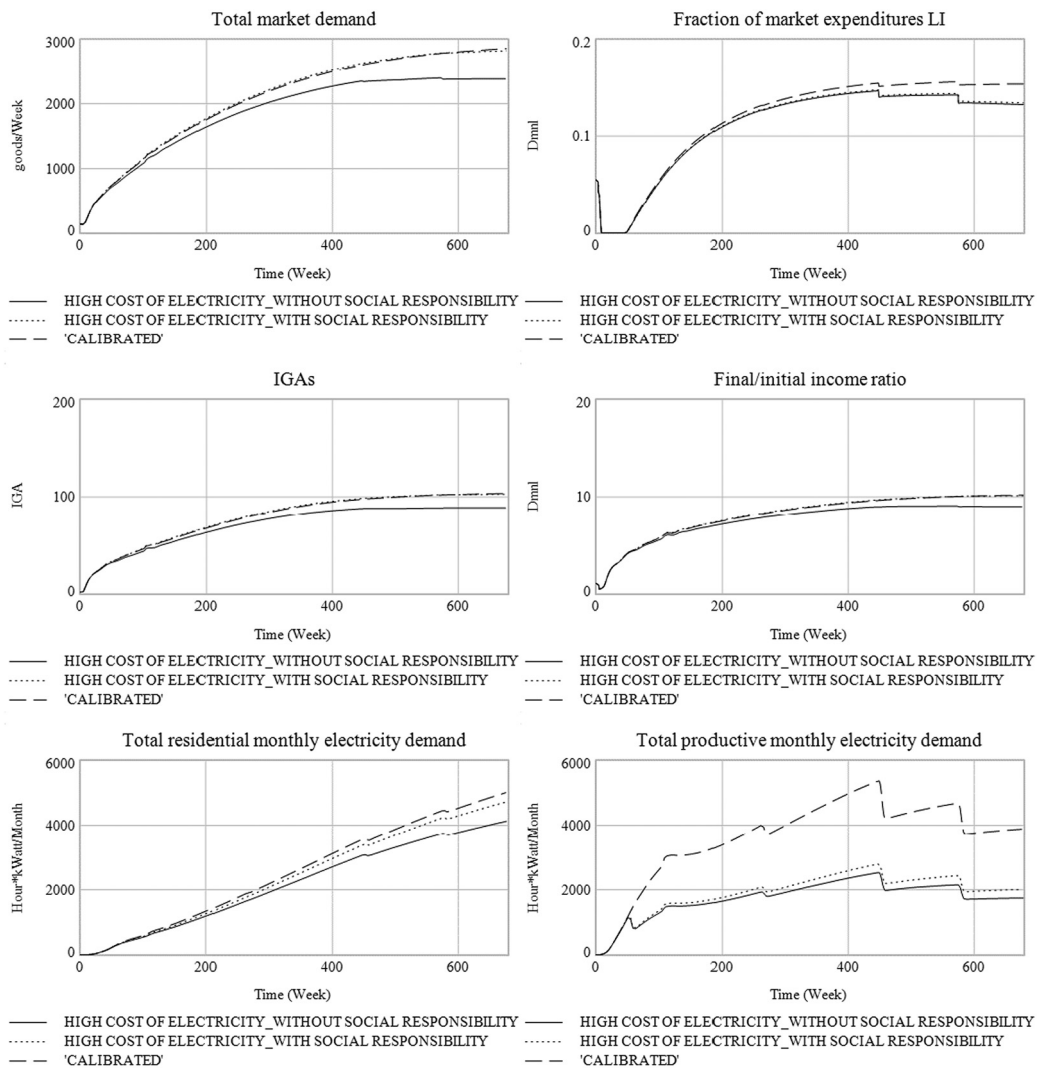


Fig. 4. Effect of electricity cost on some of the main outputs.

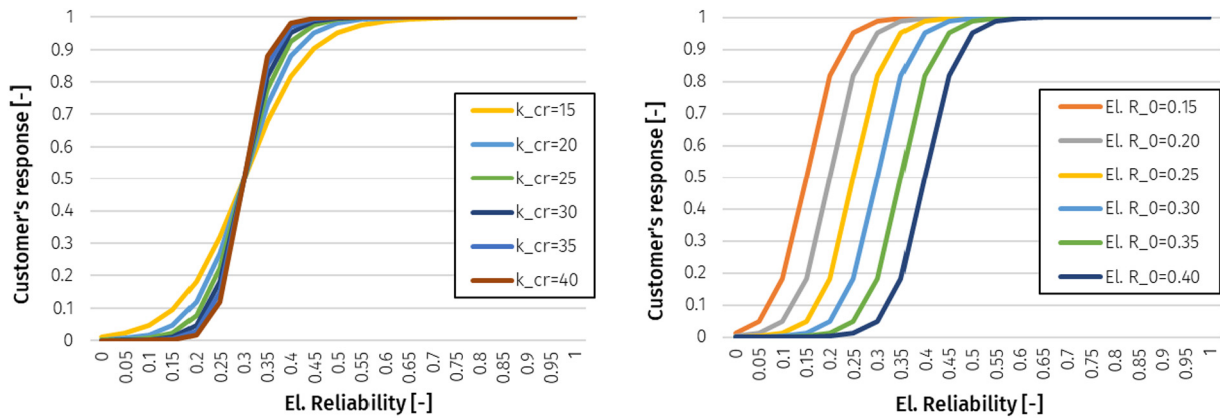


Fig. 5. Logistic function of CR variable by varying k_{cr} and keeping $R_0 = 0.3$ (left-side), and by varying R_0 and keeping $k_{cr} = 30$.

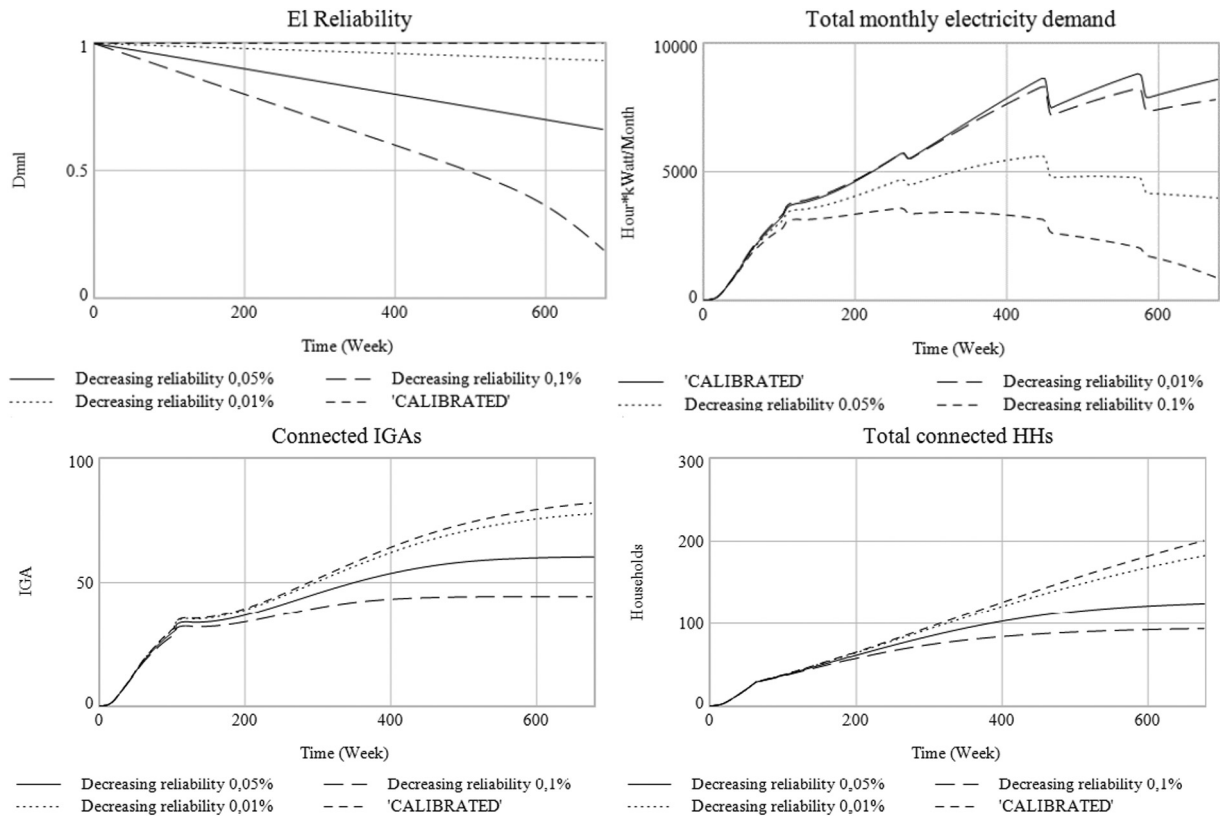


Fig. 6. Effect of electricity reliability on some of the main outputs.

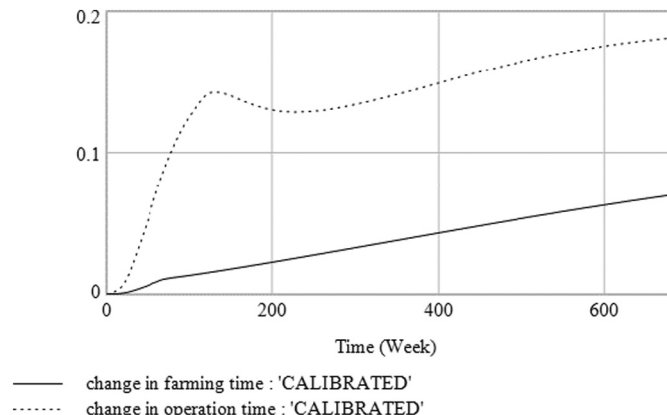


Fig. 7. Effect of electricity use on time available for working and farming [% of change].

conditions and parameter values, and to increase the knowledge of the electricity-development nexus and the dynamics behind the evolution of rural electricity demand. The goal is to derive policy implications and lessons that can be replicated for enhancing sustainable rural electrification practices in other contexts. As a novelty, the most critical and fundamental dynamics and parameters were assessed: it was found that the dynamics and initial conditions related to market operation, farming trading, population, and households' economy are the most fundamental determinants of the electricity-development nexus. From the Ikondo case, the following first set of lessons learnt to facilitate the development of more generalized electrification guidelines for sub-Saharan Africa can be derived from the identification and analysis of these critical and fundamental dynamics and parameters:

- To carry out complementary activities for improving the accessibility, and so the potential trading level, of isolated villages in order

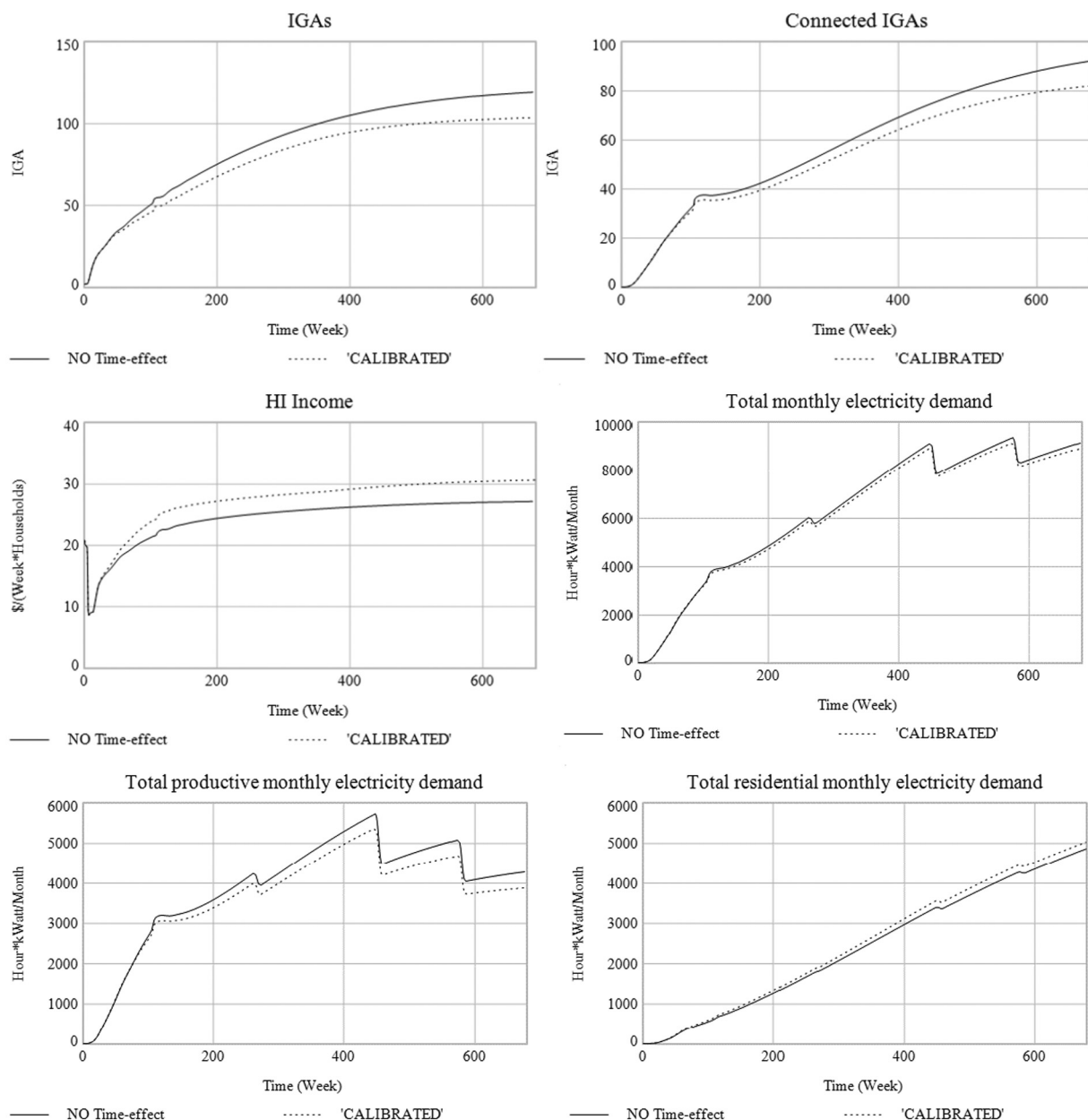


Fig. 8. Feedback of time on some of the main outputs.

to allow local people to build upon this favourable trading condition to exploit at a higher degree the benefits of having electricity available.

- To implement awareness campaigns intended to improve farming productivity for attracting external consumers and vary the level of trade of the agricultural products also after electrification, discouraging the complete leaving of agricultural activities in favour of the creation of new IGAs only.
- To innovate local production, increase market competition on new goods and services.
- To facilitate the access to resources (e.g. machines, equipment, raw materials, spares, primary goods) to allow local businesses to satisfy the local market demand.
- To implement activities and awareness campaign aimed at diversifying people's basket of goods and services: the diffusion of communication devices – e.g. TVs, radio and phones – may help, since they can introduce changes in aspirations and expenditures

of rural households for goods and services, diversifying purchases and leading people to shop locally rather than elsewhere.

- To implement socio-economic complementary initiatives aimed at reducing initial inequalities.
- To carry out capacity building activities to increase people's skills in delivering new services and products.
- In rural wide areas characterised by scattered and dispersed rural communities, to target initially the community with an initial higher number of people, in order to foster spillover effects on other smaller communities.

Future electrification programmes can stand on these results for identifying the most critical aspects to deal with in order to guarantee the long-term sustainability of energy access projects.

The interrelation between further socio-economic dynamics and electricity demand was also assessed. E.g. the effect of electricity uses on market improvements, and the related feedback, resulted to be

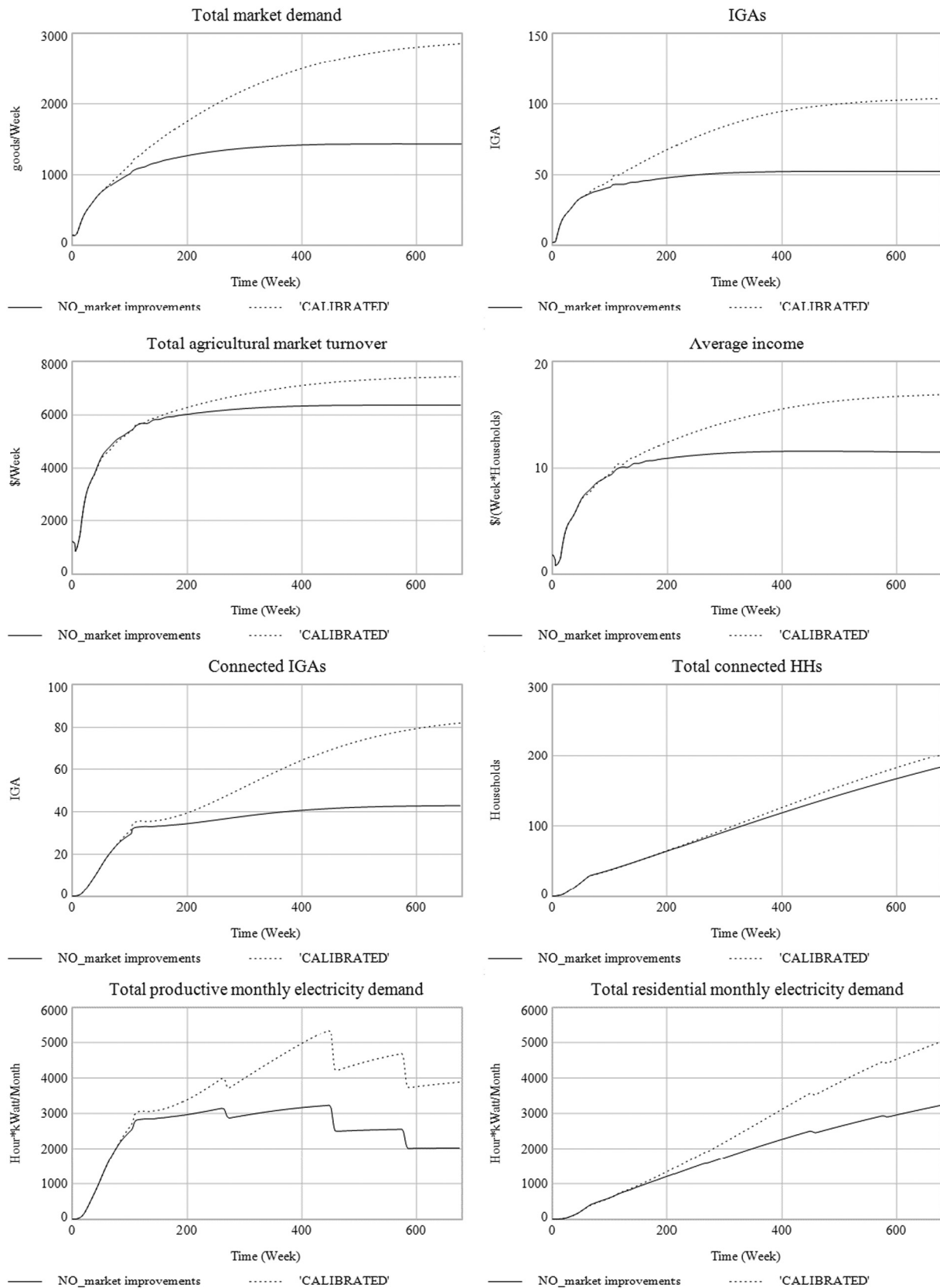


Fig. 9. Feedback of market improvements on some of the main outputs.

highly interdependent dynamics. This can have two implications: (1) from the future research and modelling point of view, the process of model simplification can start from these findings for identifying

the negligible dynamics that characterise the electricity-development nexus; (2) potential use of the model for the definition of appropriate indicators and thresholds for updating the traditional monitoring and

evaluation frameworks commonly used for assessing energy access projects.

The role of exogenous policies was also tested, providing a second set of potentially useful guidelines:

- To combine electricity access programmes with access to appropriate forms of micro-credit, in order to facilitate the creation of local IGAs.
- To ensure medium and long-term affordable and appropriate plans for electricity tariffs to people (both for the energy consumed and cost of connection) and avoid ongoing abrupt changes.
- To guarantee and maximise the highest standard of power reliability, which is an important factor to support people's willingness to pay for electricity.

These results contribute to the literature aiming at setting good practices, minimal prerequisites, and complementary activities that should be included in rural electrification projects, with potential implications on the current frameworks adopted by international organisations and donors for defining and financing energy access programmes.

This work represents the second step, after model conceptualisation and formulation, towards an increased understanding on the electricity-development nexus in rural areas. Future activities will so include model simplification and testing for different contexts than Ikondo, as well as the integration of the system dynamics model within energy optimisation tools for a more comprehensive and robust electrification planning approached.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esd.2020.10.009>.

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